Category Theory Explained — A Complete Guide

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Edited on: June 21, 2026, 6:19 AM IST

The question: What does category theory clarify about the structure of Madhyasth Darshan — and where does that notation stop?

This is a self-contained guide to describing Shri A. Nagraj's Madhyasth Darshan using **category theory**. Parts 1 to 5 introduce the ideas in plain language; Part 6 gives the precise formal theory. The ontological ground and the tier-neutral formal template are developed in *What Is Existence?* and *The Coexistence Template*; human-tier conduct and society are worked out further in *Why Humans Are Not Just Material* and *Human Behavior and Society*. Primary texts: *Madhyasth Darshan — Co-existentialism* (MVD), *Samadhanatmak Bhautikvad* (SB), and *Jeevan Vidya: An Introduction* (JV).

Category theory here is a **lens for clarity**, not a proof machine. It makes the philosophy's logic visible and shows exactly what each conclusion depends on. It does not prove the metaphysics, and it cannot supply empirical evidence.

The one big idea

Category theory is built on a single shift in attention:

> **Stop asking only "what is each thing made of?" and start asking "how does each thing relate to everything else?"**

Most of modern science explains things by **breaking them into parts** (cells, molecules, atoms, particles). Category theory instead studies the **arrows between things** — the relationships, the flows, the transformations — and treats those relationships as the real subject matter.

Madhyasth Darshan holds that **existence is coexistence**: nothing is fully understood in isolation; a thing's meaning comes from how it lives in relationship with everything else — first through **saturation** in Omnipresence (*satta*), then through definite **relationships** (*sambandh*) with other units (What Is Existence? §1.2). That two-layer picture is close to category theory's attention to structure, though saturation is not itself a morphism between units (§6.13).

```
flowchart LR
    A["Old question: what is it made of?"] --> B["Take it apart"]
    C["New question: how does it relate?"] --> D["Study the relationships"]
```

Part 1: The four basic words

Category theory has only a few core ideas. Here they are in plain language.

1. Objects = the "things"

An **object** is just a thing you want to talk about. It can be concrete (a body, a family) or abstract (justice, resolution, fulfilment). In a diagram, objects are the labelled boxes.

> *Everyday picture: the dots on a map (cities).*

2. Morphisms (arrows) = the "relationships"

A **morphism**, usually drawn as an arrow, is a relationship or a way of getting from one thing to another. It might mean "supports", "develops into", "understands", "uses rightly", or "fulfils".

> *Everyday picture: the roads between cities on the map. The roads, not the cities, are what category theory cares about most.*

3. Composition = "chaining relationships"

If there is an arrow from A to B, and another from B to C, then there is a combined arrow from A to C. This is **composition** — following one relationship by another.

> *Everyday picture: if there's a road from Delhi to Bhopal, and one from Bhopal to Amarkantak, then there is a route from Delhi to Amarkantak. You can chain them.*

In the darshan this shows up as:

```
Understanding -> Resolution -> Humane conduct -> Social order
```

Chaining these gives the shortcut "Understanding leads (eventually) to social order." The claim that the chain holds together is the interesting part.

4. Identity = "staying yourself"

Every object has a trivial "do nothing" arrow to itself, called the **identity**. It sounds empty, but it lets us ask a sharp question later: *has a person actually changed, or only stayed the* *same while looking different?*

> *Everyday picture: staying in the same city.*

That is the entire foundation. Boxes, arrows, chaining arrows, and a do-nothing arrow. Everything else is built from these.

Part 2: A few more tools, each in one breath

These are the slightly fancier tools the formal theory uses. Each is given here as a plain idea plus how the darshan uses it. You do not need to memorise them; refer back as they appear.

Tool	Plain-language meaning	Everyday analogy	How Madhyasth Darshan uses it
Functor	A faithful translation from one world of things-and-arrows to another that keeps the structure intact	Translating a recipe to another language so the steps still work	Turning relationships into the values they should fulfil
Forgetful functor	A translation that deliberately throws away some information	Photocopying a colour photo in black and white	Describing a human using only physics, dropping values and `jeevan`
Natural transformation	A consistent, across-the-board upgrade from one way of doing things to another	Upgrading every road on the map to a highway, uniformly	Shifting from “consumption” to “right-use” everywhere at once
Retract	A part that sits inside a whole, where the whole cannot be rebuilt from the part alone	A thumbnail made from a photo: you can shrink, but not un-shrink	Comfort is a real part of fulfilment, but not the whole of it
Colimit (gluing)	Joining many small pieces into one big consistent whole, along what they share	Gluing map tiles into one map where the edges match	Families joining into one undivided society
Enrichment	Recording not just “is there a relationship” but “of what quality/grade”	A road map that also marks each road’s quality	Graded values — object, established, civic — and kinds of satisfaction

Tool	Plain-language meaning	Everyday analogy	How Madhyasth Darshan uses it
Indexed enrichment / fibration	Structure that varies with who is acting — not one fixed grade for everyone	Same recipe, but what you can actually cook depends on your kitchen	Fulfilment modulated by capacity, ability, and receptivity (§6.6.1)
Mixture vs compound	Two ways of joining units: side-by-side aggregation, or fusion into a new whole	Fruit salad vs baking a cake	*Mishran* keeps each conduct; *yaugik* creates a new tier (§6.9)

Part 3: The philosophy, redrawn as a map

3.1 The four orders as a ladder

Madhyasth Darshan names four real developmental plateaus — material (*padarth*), biological (*pran*), animal (*jeev*), and knowledge (*gyan*) — each containing the ones below:

```
flowchart LR
    M["Material: rocks, water, air"] --> B["Plant: living, growing"]
    B --> A["Animal: feeling, instinct"]
    A --> K["Human: knowing, valuing"]
```

The arrows point one way for a reason. A human contains and depends on the material, plant, and animal levels — but you **cannot go backwards** and rebuild human knowing out of pure chemistry. In category-theory language this one-directional ladder (a "partial order") is the cleanest way to say:

> *Higher includes lower, but higher is not reducible to lower.*

The formalism does not *prove* this; it *records* it cleanly, so the claim is explicit and cannot be smuggled in or out unnoticed.

3.2 A human being: an actor and an instrument

The darshan says a human is **body** + `jeevan` (the sentient self), and crucially that these are not equal partners: `jeevan` is the actor, the body is its instrument. *Jeevan* is a **constitutionally complete composite atom** (*gathanpurna parmanu*) — a self-maintaining sentient unit, not an elementary particle of physics (What Is Existence? §1.6).

A natural first guess is to write this as a simple pair, "Human = Body and Jeevan." That turns out to be the **wrong** picture, because a "pair" suggests two equal, separable halves. A better plain-language picture is:

> `Jeevan` *is like a musician; the body is like the instrument. The music (values, evaluation, resolution) comes through the instrument but is not produced by the wood and strings alone. And you cannot swap them — the instrument does not play the musician.*

```
flowchart TD
    J["Jeevan: the actor (values, evaluation)"] --> H["Human (the music that res
    Bd["Body: the instrument (food, sleep, health)"] --> H
```

This asymmetry is the heart of the darshan's claim that a human cannot be fully studied as a body alone. In category-theory terms, the "black-and-white photocopy" (the forgetful functor that keeps only physics) loses the musician and keeps only the instrument.

3.3 Delusion: mistaking a part for the whole

- **Comfort** (pleasure, wealth, health) is a genuine *part* of human fulfilment.
- But **fulfilment** is bigger than comfort; you cannot rebuild full fulfilment out of comfort alone.

In the tools table this is a **retract**: comfort fits inside fulfilment, but the arrow does not reverse. **Delusion**, in Madhyasth Darshan, is precisely the error of treating the part as if it were the whole — believing "maximise comfort" equals "achieve fulfilment". The texts diagnose a related pathology as unbounded "more" (JV p. 41); the darshan moves toward **definite completeness**, not open-ended maximisation (The Coexistence Template §5).

```
flowchart LR
    C["Comfort (a real part)"] --> F["Fulfilment (the whole)"]
    F --> C
```

Both arrows exist (comfort feeds fulfilment; fulfilment includes comfort), but going out and coming back does **not** return you to where you started — something is always left over. That "left over" is exactly what the darshan calls resolution, the part comfort can never supply.

3.4 Right-use: an all-or-nothing upgrade

The darshan contrasts **consumption** (using nature as raw material to be extracted) with **right-use** (using it complementarily, sustainably).

Right-use must be **consistent across every domain at once**. You cannot claim to practise right-use toward the forest while exploiting workers, or right-use toward workers while poisoning rivers. A genuine shift to right-use upgrades *every* relationship uniformly, the way upgrading a road network only counts if you upgrade the whole network, not one favourite road.

```
flowchart LR
    Con["Consumption everywhere"] -->|"uniform upgrade"| Right["Right-use everyw
```

So “partial right-use” is, structurally, not yet right-use. That matches the darshan’s claim that selective ethics is not yet humane conduct.

3.5 Society: gluing families that agree — and transmitting understanding

How does a good society form? Not by force, and not by simply piling up individuals. The darshan says it grows from families and communities into one “undivided society”.

The “gluing” tool gives a precise condition for when this works:

> *Many families can be glued into one society* **exactly when they agree about the people and values they share.** *Where two families place contradictory demands on the same shared person, the gluing is forced to collapse or break.*

```
flowchart TD
    Shared["Shared members / shared values"] --> F1["Family 1"]
    Shared --> F2["Family 2"]
    F1 --> S["Undivided society (clean glue)"]
    F2 --> S
```

Gluing alone is not enough at the knowledge order. An assembly **persists** while its relationships are fulfilled (natural state) and **declines** when they are not (excited state); and every persisting assembly **transmits its method of composition** across generations — by education-*sanskar* at the human tier (The Coexistence Template L4–L5). A society that coheres locally but fails to transmit understanding decays on member turnover.

Part 4: How this formalism helps

Why bother translating a philosophy into this language at all? Six concrete payoffs.

1. **It forces clarity.** You cannot draw an arrow without saying what relates to what. Vague claims (“everything is connected”) become specific (“this supports that; that fulfils the other”). Fuzzy ideas either sharpen or fall apart.
2. **It exposes hidden assumptions.** Each conclusion, drawn as a chain of arrows, makes visible exactly which link is doing the work. For example, “a human is not just a body” turns out to rest entirely on one assumption: that two physically identical acts can

differ in value. The formalism puts a spotlight on that assumption instead of letting it hide.

3. **It catches inconsistencies.** Drawing a diagram reveals when two paths that *should* agree actually don't. The "delusion" idea is literally a diagram that fails to close up — a precise picture of a confused belief.
4. **It enables fair comparison.** Once materialism, religion, and Madhyasth Darshan are each drawn as "worlds of things and arrows", you can compare what each one keeps and what each one throws away. Reductionism becomes "the translation that forgets values" — a clear, checkable description rather than an insult.
5. **It is teachable and language-neutral.** Diagrams cross language barriers. A student who finds the Sanskrit-derived vocabulary daunting can still follow boxes and arrows, then attach the names afterward.
6. **It builds a bridge to science and computation.** The same mathematics is used in physics, computer science, and AI. Phrasing the darshan in it opens a door for dialogue with those fields instead of leaving philosophy and science speaking different languages.

```
flowchart TD
    Phil["Philosophy in words"] --> Cat["Same ideas as boxes and arrows"]
    Cat --> Clear["Clarity"]
    Cat --> Hidden["Hidden assumptions exposed"]
    Cat --> Compare["Fair comparison"]
    Cat --> Teach["Easier teaching"]
    Cat --> Bridge["Bridge to science"]
```

Part 5: How it can actually be used

This is not only an academic exercise. Here are practical applications, aligned with the template's forward and reverse uses (The Coexistence Template §1.1).

In education

Teach values as **relationships to be fulfilled** rather than rules to be obeyed. A curriculum can map each relationship (parent-child, teacher-student, citizen-society) to the values it carries, and ask students where the arrows are currently broken.

As an ethics or technology checklist

Before adopting a technology or policy, ask the "right-use" question structurally: - Does this upgrade *every* affected relationship, or only some while harming others? - Can its benefits be "lifted back" into coexistence, or do they depend on extraction somewhere?

A tool passes only if the upgrade is uniform — a concrete, auditable test (naturality of §6.7).

In policy and social design

Use the gluing condition as a diagnostic: where a community is failing to cohere, look for **shared people or resources carrying contradictory expectations**. Add transmission: does the assembly institutionalise understanding for incoming members, or only rules?

In AI and value alignment

Modern AI struggles with exactly the darshan's complaint: optimising a single number (engagement, profit, "reward") maximises the wrong thing. Graded fulfilment (enrichment over a value preorder, §6.6) is an argument for **multi-level objectives** instead of one scalar reward.

In interfaith and science-philosophy dialogue

Because the formalism states what each worldview *keeps and forgets*, it lets very different traditions compare notes without insult or conversion.

As a research programme

Open questions include: how saturation in O relates to the categories of unit-to-unit structure; whether transmission τ admits a clean coalgebra or indexed-category model; and whether complementarity of need (template L3) can be fully internalised or must remain an external guard on admissible diagrams (§6.9.2).

Part 6: The complete formal theory

Parts 1 to 5 gave the intuition. This part gives the precise version for readers comfortable with (or curious about) the mathematics, and states exactly what the formalism can and cannot carry.

6.0 Design discipline (the rules this theory obeys)

A loose use of categorical vocabulary can mislead. This theory commits to five rules.

1. **One kind of morphism per category.** A single category may not mix causal, developmental, epistemic, and normative arrows, because then composition is meaningless. Different kinds of relation live in different categories, connected by functors.
2. **Composition and identities must be specified and associative.** If a composite has no clear meaning, the structure is a quiver (a labelled directed graph), not a category, and is labelled as such.
3. **Universal properties must be stated and, where claimed, checked.** Products, colimits, adjoints, and monoidal structures are not invoked by name unless their defining property is given.

4. **Every nontrivial claim names its hidden premise.** Where a conclusion depends on a contested Madhyasth assumption, that premise is stated explicitly. The categorical step is then valid only relative to it.
5. **Propositions are conditional, not theorems about reality.** Nothing here proves `jeevan`, constitutional completeness, or coexistence. The propositions show what *follows structurally if* the premises are granted.

Category theory is **structuralist**: by the Yoneda principle, an object is determined entirely by its pattern of relations to other objects. That will matter, because Madhyasth Darshan holds that `jeevan` is a **substantial** entity, not merely a relational role — the deepest limit of the whole project (§6.13.1).

6.1 Architecture: several categories and functors, not one

Instead of a single all-purpose category, we use a small system of categories, each internally clean, related by functors, natural transformations, and — where needed — structure that sits outside ordinary inter-unit morphisms.

```
flowchart TD
    Sat["Sat: saturation slice (O not in U)"]
    Ord["Ord: poset of orders"]
    Cap["Cap: capability profiles"]
    Phys["Phys: physical descriptions"]
    Liv["Liv: living/human descriptions"]
    Rel["Rel: unit relationships (sambandh)"]
    Val["Val: values (graded preorder)"]
    Eval["Eval: evaluation (knowledge order)"]
    Conduct["Conduct: enriched over fulfilment"]
    Comp["Comp: composition (mixture / compound)"]
    Soc["Soc: social gluings (colimits)"]
    Trans["Trans: transmission across generations"]

    Sat -.->|"energises/regulates"| Liv
    Liv -->|"cap(u)"| Cap
    Cap -.->|"filters / grades phi"| Rel
    Liv -->|"U (forget jeevan)"| Phys
    Rel -->|"V (valuation)"| Val
    Rel -->|"assemble"| Soc
    Comp -->|"kappa"| Soc
    Liv -->|"mu (evaluate)"| Eval
    Eval --> Val
    Conduct -->|"grades"| Val
    Liv -->|"inhabits"| Ord
    Soc -->|"tau"| Trans
```

Sat (saturation). Omnipresence **O** is not a unit and not an object of `Rel` (The Coexistence Template §3.1). Saturation is the constitutive immersion of every unit in *O* (template A1–A2;

What Is Existence? §1.2). Categorically it is best treated as an **ambient** or **enrichment base** — a family of regulators indexed by units — not as a morphism $u_1 \to u_2$. Terminal-object and topos-as-space readings of *vyapak* remain stretches (§6.13.4).

Eval and μ. Evaluation is defined **only for knowledge-order units** (template D6): *jeevan* assesses value delivered in a relationship. Model it as an endofunctor or process `μ : Liv → Liv` (or a functor into a category of assessments) that does not factor through `Phys`. Justice is the composite **ρ → φ → μ → mutual satisfaction** — an **operator over** *V*, not a member of *V* (template D7).

Trans and τ. Transmission re-instantiates an assembly's composition method across member turnover (template L5). A clean categorical home is an **indexed category** or **coalgebra**: generations as objects, τ as structure-preserving maps carrying *rachna vidhi* / education-*sanskar* forward. Colimits alone do not generate τ (proposition P8).

Cap and cap(*u*). Fulfilment is not automatic from recognising a relationship: it is modulated by **capacity** (*kshamata*), **ability** (*yogyata*), and **receptivity** (*patrata*) (template D5a). Model cap as a functor `cap : Liv → Cap` into a category of capability profiles; `Rel` and `Conduct` are then **fibred** or **indexed** over unit capability (§6.6.1).

Comp and κ. Composition takes two modes — **mixture** and **compound** (template D8). Only **compound-mode** κ creates a new tier; mixture aggregates at the same tier. Categorically this is the difference between a **coproduct preserved at fixed order** and a **colimit that changes signature** (§6.9).

6.1.1 Relation to the coexistence template

The Coexistence Template states the tier-neutral structure formally; this paper supplies natural notation for parts of it. The mapping is intentional, not a proof that the symbols coincide.

Template	Symbol / law	Categorical analogue in this paper	Fit / gap
Medium, immersion	**O**, A1–A2	Ambient `Sat` ; not in `Rel`	Medium is constitutive; CT has no default for non-morphismic ground
Units	**U**, D1	Objects across categories	Strong
Relationships	**R**, D3	Category `Rel` ; expectation profiles **E(r)** on arrows	Strong; association vs relationship is typing, not composition
Values	**V**, D4	Preorder `Val` ; enrichment base **W**	Moderate — graded taxonomy (object / established / civic) compresses in §6.6

Template	Symbol / law	Categorical analogue in this paper	Fit / gap
Recognition, fulfilment	**ρ**, **φ**, D5	Morphisms/processes on `Rel` / `Conduct`; fibred over `Cap`	Moderate — cap(u) = ⟨ksh, yog, pat⟩ in §6.6.1
Evaluation	**μ**, D6	`Eval` on `Liv` only	Moderate — order-restriction is essential (P10)
Composition	**κ**, D8	`Comp`: mixture = coproduct; compound = tier-changing colimit	Moderate — modes distinguished in §6.9
Complementarity	**L3**	—	**Gap:** selection rule for which diagrams are admissible; CT is silent (Template §7)
Persistence	**L4**, D9	Colimit compatibility + natural/excited typing	Partial
Transmission	**τ**, L5	`Trans` coalgebra / indexed maps	Weak — not derivable from κ alone
Tier iteration	**L6**	Poset `Ord` + iterated colimits	Strong
Four orders	D2	Poset `M ≤ B ≤ A ≤ K`	Strong

The template's §7 comparison row states the division of labour plainly: category theory is the natural notation for **κ** and **L6**, but **L3** (complementary need as the engine of assembly) is the selection rule the notation lacks. Any colimit model of society must be supplemented by an external criterion — complementary deficiency and surplus — for which diagrams count as development-progression bonding rather than arbitrary gluing.

6.2 The poset of orders (mereological containment)

The four orders are modelled by a **thin category** (a partial order), with at most one arrow between any two objects.

```
Objects:   M, B, A, K  (material, biological, animal, knowledge)
Order:     M <= B <= A <= K
Reading of x <= y:  "order y contains and depends upon order x"
```

- **Morphism-kind:** containment/dependence only.
- **Composition:** transitivity of `<=`. Trivially associative.
- **Identities:** reflexivity `x <= x`.

The claim "the higher-order universe contains the lower-order universe" (MVD Ch. 3) is **mereological** (part/whole), not transformational. Posets are the natural home of part/whole structure.

The non-existence of `K <= M` is exactly the anti-reductionist content; here it is a flat structural fact, not an argument. The poset *encodes* the claim, it does not *prove* it.

6.3 Reduction as a forgetful functor

Let `Liv` be human descriptions that include value-bearing structure, and `Phys` purely physical descriptions. Define the forgetful functor:

```
U : Liv -> Phys
U(human situation) = its physical substrate
U(humane transition) = the underlying physical change
```

Is `U` faithful? (`U` is faithful iff distinct `Liv`-morphisms never collapse to the same `Phys`-morphism.)

- **Premise (stated):** two acts can be physically identical yet differ in value — e.g. genuine justice versus an outwardly identical imitation differ in `jeevan`-level content (SB Ch. 7).
- **Conditional conclusion:** *if* that premise holds, then `U` **is not faithful**. The functor formalism contributes precision, not evidence; the premise does all the work.

Does `U` have a left adjoint (a "free jeevan" functor)? A left adjoint would freely generate life/value from bare matter.

- **Premise (stated):** *gathanpurnata* is reached by irreversible **Development Progression** (*vikas-kram*) through effort–motion–result (*shram–gati–parinam*), not a free, uniform construction ([What Is Existence?](#) §1.3, §1.6).
- **Conditional conclusion:** *if* development is selective and irreversible, there is **no left adjoint with iso unit**; "adding life freely to matter" is not a structural operation.

What this does not do. It does not show the premise is true. A physicalist who denies the value-distinct premise keeps `U` faithful and is untouched.

6.4 The human: not a product, but an action

A categorical **product** (`Human = Body x Jeevan`) is wrong here: a product is symmetric and freely separable, but the darshan says body and `jeevan` are inseparable in function and asymmetric ("`jeevan` is the bearer; the body is a vehicle and medium", SB Ch. 7, JV Ch. 1). A better model: `jeevan` as a **monoid acting on body-states**.

```
Let (J, *, e) be a monoid of jeevan-activities (valuing, evaluating, resolving).
Let Bdy be the body-states.
A human is an action:   act : J x Bdy -> Bdy
with                    act(e, b) = b
                        act(j1 * j2, b) = act(j1, act(j2, b))
```

This captures asymmetry (`J` acts on `Bdy`, not vice versa), inseparability-in-function (a human is the *action*, not a detachable pair), and "results beyond bodily need" (the orbit can contain states unreachable by body-dynamics alone).

This is a modeling choice, not the unique one; a fibration, comma object, or monad algebra would each capture part of the same asymmetry. The non-uniqueness is itself an issue (§6.14).

6.5 Delusion as a retract that is mistaken for an isomorphism

Let `F` be **complete fulfilment** and `C` **bodily comfort**. Comfort is genuinely part of fulfilment:

```
i : C -> F      (comfort contributes into fulfilment)
p : F -> C      (fulfilment includes a recoverable comfort aspect)
with    p . i = id_C     (C is a retract of F)
but     i . p != id_F    (F cannot be rebuilt from C alone)
```

So `C` is a **retract** of `F` but **not isomorphic** to it.

Definition (delusion). Delusion is the false assertion that `i` is an isomorphism, i.e. treating `i . p = id_F` — identifying the two distinct endomorphisms `id_F` and `i . p` of `F`. Equivalently: treating movement toward unbounded "more" as movement toward completeness (Template §5).

```
flowchart LR
    C["C: comfort"] -->|i| F["F: fulfilment"]
    F -->|p| C
```

Because both `id_F` and `i . p` have the same domain and codomain `F`, asking whether they are equal is well-typed; the darshan asserts they are unequal. This is a clean structural statement of "pleasure/wealth/health are necessary but not sufficient" (MVD Ch. 4).

6.6 Fulfilment as enrichment, not a scalar

The texts give value a **graded taxonomy** (Template D4): object value at all orders; *jeevan* values and established relationship values (trust, respect, affection, …) at the knowledge order; civic values in assembly tiers. MVD Ch. 4 also distinguishes sensory (momentary), intellectual (lasting), and existential (non-transformable) satisfaction. Model the ordering by **enrichment over a preorder**.

```
Let W = (sensory <= intellectual <= existential), made monoidal by min, unit = e
Enrich the conduct category over W:
the hom-object Conduct(f, g) is an element of W recording the quality of fulfilm
```

Enriched composition requires `Conduct(g,h) (x) Conduct(f,g) <= Conduct(f,h)` — "a chain is only as high-grade as its weakest link." This refuses the collapse of fulfilment into one additive scalar — the Madhyasth objection to utility/happiness maximisation — and aligns with definite completeness rather than open-ended optimisation.

The base `W` and the choice of `min` are modeling decisions. The enrichment captures *ordering of qualities* well; it does not capture *what existential realisation is*, nor the full D4 class hierarchy without a finer indexed enrichment. **Fulfilment capacity** (template D5a) supplies that finer index — §6.6.1.

6.6.1 Fulfilment modulated by capacity (cap(u))

Template D5a writes **cap(u) = ⟨ksh, yog, pat⟩** for capacity (*kshamata*), ability (*yogyata*), and receptivity (*patrata*) (MVD p. 62). Recognition ρ and fulfilment φ are universal across orders (template L1), but **φ is realised only at the level cap(u) permits** — not by receptivity alone. In the material order, **ascending** (*agreshan*) is balance and receptivity gained while converting capacity and ability into effort (*shram*); **frustration** (*kshobh*) is the shortcoming in that conversion — "incompleteness of receptivity" (MVD p. 79). In the knowledge order, cap(u) for worldview arises from environment, study, and prior *sanskar* (MVD p. 134); extent of receptivity is qualification (*arhta*), yielding perspective (*drishti*) and worldview (*darshan*) (MVD p. 142) (What Is Existence? §1.2.2).

The capability category `Cap`

Define a category `Cap` whose objects are capability profiles **c = ⟨ksh, yog, pat⟩**, each coordinate living in an order-specific preorder (material profiles are not human profiles). A morphism **c** → **c′** means coordinate-wise improvement — more scope, more convertibility, more readiness.

Each unit carries a profile by a functor:

```
cap : Liv -> Cap
cap(u) = <ksh(u), yog(u), pat(u)>
```

`Phys` has a trivial or collapsed image: lower-order units convert capacity and ability into bonding automatically; the interesting fibred structure appears at the knowledge order, where cap can **fail** to improve and where μ can misfire (over-, under-, mis-evaluation, MVD p. 38).

Three roles of the triad (what each coordinate gates)

Coordinate	Textual role	Categorical gate
pat (*patrata*)	Readiness for a relationship's expectations and values to register	**Existence of morphisms:** `(u,r) ∈ Rel` only if `pat(u) ≥ pat_req(r)`
yog (*yogyata*)	Competence to convert capacity into effort, conduct, or bonding	**Composition of φ-steps:** the composite $\varphi_2 \circ \varphi_1$ is defined only if yog(u) suffices along the chain; failure = *kshobh*
ksh (*kshamata*)	Scope for participating in relationships and bearing what coexistence makes available	**Grade ceiling in *W*:** effective satisfaction grade is `min(W_chain, ksh_ceiling(u))`

Receptivity alone does not produce fulfilment: a unit may **recognise** a relationship (ρ) yet deliver only a truncated value flow because yog or ksh is insufficient — the template's diagnosis of frustrated or partial order.

Fibred enrichment over units

Combine §6.6's enrichment base **W** with **Cap** by passing to a **fibred category** (or indexed category) over units:

```
pi : E -> Rel
```

For each relationship arrow `r : u -> v` in `Rel`, the fibre carries **Conduct**-morphisms enriched in **W**, but:

1. The arrow `r` appears in the fibre over `u` only if **pat(*u*)** meets the relationship's requirement.
2. Composition in the fibre uses **min** on **W** as in §6.6, and is **defined** only when **yog(*u*)** (and yog(v) where needed) supports the composite effort.
3. The grade recorded in the fibre is capped by **ksh(*u*)**.

Effective fulfilment along `r` is therefore not a single functor `Rel → Val` but a **family** `ϕ_u` indexed by units, with:

```
grade(ϕ_u(r)) = min(grade_inherent(r), ksh(u))   in W
```

when yog(u) and pat(u) permit φ at all. At the knowledge order, **μ** compares `grade(ϕ_u(r))` to the value inherent in the relationship — the three failure modes are precisely μ operating on a φ that was already cap-truncated or cap-blocked.

Justice as a cap-sensitive composite

Justice (template D7) is **ρ → φ → μ → mutual satisfaction**, not a member of V. Categorically it is a **partial composite** on the fibred conduct category:

```
Justice_u(r) = mu_u o phi_u o rho_u(r)
```

defined only when all three coordinates of cap(u) permit the chain. The knowledge-order discontinuity (template P3) is visible here: in the first three orders cap is high enough that the composite executes definitely; in the knowledge order the same diagram may **fail to compose** — which is exactly why human assembly is the unfinished tier.

What this does not do

The preorders on ksh, yog, and pat are **order-specific** and textually grounded, not derived from category theory. The fibred model **organises** D5a; it does not prove that cap(u) has the stated coordinates or that frustration is always "incompleteness of receptivity." A simpler stopgap — a single scalar "competence" per unit — would lose the template's distinction between recognising, converting, and bearing.

6.7 Right-use as a natural transformation

Let `D` be "domains of use" (nature, wealth, body, knowledge) and `Pr` a category of practices. Define two functors:

```
C : D -> Pr     consumption practice
R : D -> Pr     right-use practice
```

A natural transformation `eta : C => R` has components `eta_X : C(X) -> R(X)` such that for every domain-morphism `f : X -> Y`:

```
R(f) . eta_X  =  eta_Y . C(f)        (naturality square commutes)
```

```
flowchart LR
    CX["C(X)"] -->|"eta_X"| RX["R(X)"]
    CX -->|"C(f)"| CY["C(Y)"]
    RX -->|"R(f)"| RY["R(Y)"]
    CY -->|"eta_Y"| RY
```

The naturality square is the substantive content: the shift from consumption to right-use is **uniform across all domains**. **Conditional conclusion:** *if* right-use is one principle (not domain-by-domain opportunism), then partial right-use that violates naturality is incoherent — a precise version of "selective ethics is not yet humane conduct."

6.8 Undivided society as a colimit — with persistence and transmission

```
Index category J:
  objects    = families and communities
  morphisms = inclusions of shared members / shared relationships
Diagram   D : J -> Rel
  assigns each family its value-structure, and each overlap the shared sub-struc
Undivided society := colim D     (the gluing of all families along shared member
```

The simplest case is a pushout of two families sharing a member:

```
flowchart TD
    O["overlap: shared member"] --> F1["family 1"]
    O --> F2["family 2"]
    F1 --> S["colimit: undivided society"]
    F2 --> S
```

Universal property. Any consistent assignment of value-fulfilment to all families that **agrees on overlaps** factors *uniquely* through `colim D`.

Compatibility (template L4). The colimit behaves well **only if the diagram is compatible** — families assign the *same* values to shared members. Conflicting assignments force a **quotient** that collapses distinctions or degenerates. Natural state ↔ fulfilled relationships; excited state ↔ decomposition pressure.

```
Compatible local value-structures  ->  clean gluing  ->  undivided society
Conflicting local value-structures ->  forced collapse  ->  no coherent undivide
```

Transmission (template L5, P3–P4). At the knowledge order, sustainment also requires **τ**: common cause, goal, and programme (MVD p. 55), carried by education-*sanskar*. A colimit that glues families in one generation but lacks τ across turnover is structurally incomplete — the template predicts decay when understanding is not transmitted, even if local φ held momentarily.

This is where the mathematics contributes a genuine, independent result about gluing: **universal order is coherent exactly when local value-structures agree on what they share.** Transmission is a further clause category theory does not derive from colimits alone.

At the knowledge tier, a successful **compound** κ (§6.9) — not merely mixture — is what produces a new social conduct with its own sig(·); the colimit in `Soc` should be read as **compound-mode** gluing when the assembly is meant to ascend a tier.

6.9 Composition: mixture and compound (κ)

Template D8 distinguishes two modes of composition κ (MVD p. 42; What Is Existence? §1.2.2):

- **Mixture** (*mishran*): components **all maintain their respective conducts** — aggregation without a new joint conduct.
- **Compound** (*yaugik*): components combine **in definite proportion**, **discard their own conducts**, and **present another kind of conduct** — a genuinely new unit with its own **sig(*u*) = ⟨roop, gun, svabhav, dharma⟩**.

Only compound-mode κ creates a new tier of the hierarchy (template L6). Development Progression (*vikas-kram*) toward *gathanpurnata* is the canonical **compound** path (MVD p. 8; SB pp. 55, 59). Awakening Progression (*jagriti-kram*) runs in already constitutionally complete *jeevan* and must not be conflated with material-tier κ (What Is Existence? §1.5).

Categorical semantics: two colimit-like constructors

Work in a category `Unit` of units typed by order (objects carry a label in `Ord`).

Mixture κ_mix(u_1, u_2): a **coproduct** (or coproduct preserved in a subcategory at **fixed order**)

```
u1 --i1--> u1 sqcup u2 <--i2-- u2
```

with injections $\mathbf{i_1}$, $\mathbf{i_2}$, such that:

- Each component's conduct factorises through its injection — **no fusion** of *dharma*.
- sig(κ_mix) **decomposes** as the pair (sig(u_1), sig(u_2)) — same tier, side-by-side.
- Example gloss: co-present units in a mixture without a new joint law — aggregate, not ascent.

Compound κ_comp(u_1, u_2): a **colimit** (often a **pushout** or **coequaliser** following a bonding span) that **quotients** the coproduct by a **fusion** relation

```
         span of bonding (complementary need)
u1 --------------------> u_new <-------------------- u2
```

such that:

- The old separate conducts are **not** recoverable as independent projections — they are **identified** in the new conduct.
- sig(κ_comp) is a **new** signature; the output may occupy the **same or a higher** tier (hungry/overfull bond → molecule; bond chain → *gathanpurna* threshold).
- **L6 iteration** applies to κ_comp outputs, not to κ_mix outputs.

```
flowchart TD
    subgraph mix ["Mixture (mishran)"]
        M1["u1"] --> Mplus["u1 + u2"]
        M2["u2"] --> Mplus
    end
    subgraph comp ["Compound (yaugik)"]
        C1["u1"] --> Cnew["u_new: new conduct"]
        C2["u2"] --> Cnew
    end
```

Relation to §6.8 colimits

§6.8 models **undivided society** as `colim D` over families. Read by mode:

Reading	Mode	Outcome
Families cohabit but retain separate household conducts	**Mixture**	Coproduct-like diagram; no new social *dharma*
Families fuse into one **undivided** conduct with shared values on overlaps	**Compound**	Colimit with compatibility (L4); new tier of assembly

The **universal property** of §6.8 is the **compound** case: agreement on overlaps is the fusion relation. Mixture corresponds to a **disjoint coproduct** of family categories with no shared quotient — weaker cohesion.

Order-relative examples

Tier	Mixture-like	Compound-like
Material	Co-present atoms without molecular bond	Hungry/overfull atoms → molecule; progression toward *gathanpurna*
Biological	Cells co-located without multicellular integration	Cells → organism with new joint conduct (JV p. 82)
Knowledge	Individuals in proximity without family order	Family → community → undivided society (MVD p. 55)

6.9.1 Complementarity as an external guard (L3)

Template **L3** drives κ_comp: bonding from **complementary deficiency and surplus** (MVD p. 8), not from arbitrary pairs. Category theory supplies **the form** of colimits and coproducts; it does **not** supply **which spans are admissible**.

Model L3 as a **subcategory** or **predicate** on spans:

```
AdmissibleComp ⊂ Span(Unit)
(u1 <- s -> u2) admissible  iff  need(u1) complements surplus(u2) [and order rul
```

Only admissible spans define κ_comp transitions. This is proposition **P9** restated at the composition layer: **L3 is a selection rule on diagrams**, not a universal property. The Petri net of §6.10 labels **transitions**, not **which transitions nature permits**.

Conditional conclusion: *if* L3 is the engine of assembly, then a categorical model of Madhyasth Darshan must treat `AdmissibleComp` (or an equivalent guard) as **given data** alongside the categories — the same division of labour noted in The Coexistence Template §7.

6.10 Development as a monoidal/Petri process

Development Progression (*vikas-kram*) — distinct from **Awakening Progression** (*jagriti-kram*) in already constitutionally complete *jeevan* (What Is Existence? §1.2, §1.5) — runs through effort–motion–result (*shram–gati–parinam*). *Kaal* is the duration of that activity, not an independent container (Nature of Time §1.1).

Hungry/overfull atomic bonding is **compound-mode** κ_comp guarded by L3 (§6.9.1). The progression is **resource-sensitive**, best modelled by a **symmetric monoidal category** (equivalently the free such category on a Petri net), where the tensor is co-presence of resources and **transitions consume inputs to produce outputs**.

```
Places (resources):   matter, effort, partial composites, gathanpurna threshold
Transitions (kappa_comp only; each guarded by AdmissibleComp):
   bond        : hungry-atom (x) overfull-atom  -> molecular composite   [compou
   complete    : composite (x) effort           -> gathanpurna parmanu   [compo
```

Mixture transitions (if included) are **identity-preserving** on each factor — token duplication without fusion — and do **not** advance Development Progression toward a new tier.

Transitions toward *gathanpurnata* are generally **not invertible** — the constitutionally complete atom does not revert to insentient configuration (SB p. 55). Insentient cycles may still exhibit **cyclical restoration** in exchange (SB pp. 52–53; Nature of Time §3.5). The Petri model captures **bookkeeping** of compound κ along *vikas-kram*, not metaphysical content, and would equally model "matter + effort → magic." Structure cannot certify content.

6.11 Propositions (conditional, with hidden premises exposed)

These are stated as: **structural claim, given premise P**. None is a theorem about reality.

#	Structural claim	Required premise (the contested part)	Status
P1	`U : Liv -> Phys` is not faithful	Two physically identical acts can differ in value	Valid given premise; premise unproven

#	Structural claim	Required premise (the contested part)	Status
P2	No left adjoint `F ⊣ U` with iso unit	Development Progression is selective/irreversible, not free	Valid given premise; premise unproven
P3	Comfort `C` is a retract of fulfilment `F`, not iso	Sensory satisfaction is a proper part of fulfilment	Valid given premise; premise plausible
P4	Delusion = asserting `i . p = id_F`	P3 holds	Coherent definition
P5	Right-use is a natural transformation; partial right-use breaks naturality	Right-use is a single cross-domain principle	Valid given premise
P6	Undivided society = `colim D` exists cleanly iff diagram is compatible	Shared members carry consistent values	Close to a real theorem about colimits
P7	Means-only categories cannot generate value-morphisms	The is/ought gap (Hume), not Madhyasth-specific	Valid; philosophy supplies it
P8	Transmission τ is not reconstructible from colimits / κ alone	L5 is independent of L3–L6 gluing	Valid; τ needs separate formalisation
P9	Complementarity (L3) is not derivable from universal properties	Assembly is driven by need/surplus, not any colimit	Valid; external selection rule on diagrams
P10	Evaluation μ does not factor through `U` or lower-order categories	μ is jeevan-only (D6)	Valid; encodes knowledge-order discontinuity
P11	Effective φ is fibred over cap(u); justice is a partial composite	D5a: fulfilment modulated by ksh, yog, pat	Valid given premise; preorders not derived
P12	κ_mix (coproduct) ≠ κ_comp (fusion colimit); only κ_comp iterates L6	D8: mishran vs yaugik	Valid; distinguishes §6.8 readings

P6 is the only place ordinary category theory does substantive independent work on gluing. Elsewhere the notation sharpens and exposes the argument, but the load is carried by a Madhyasth premise or by structure outside CT (P8–P9, L3 guard in §6.9.1).

6.12 Coverage map: what is represented, and how well

Madhyasth concept	Categorical representation	Fit	Comment
Saturation in O (A1–A2)	Ambient `Sat`; not `Rel` morphism	Weak–moderate	Two-layer ontology; CT has no native ground
Four orders contain lower orders	Poset `Ord`	Strong	Mereology is what posets are for
Anti-reductionism	No `K <= M`; non-faithful `U`	Strong structurally	Encodes, does not prove
Gathanpurna parmanu	Irreversible transition in Petri/monoidal layer	Moderate	Bookkeeping only
Body vs `jeevan`, asymmetry	Monoid action	Moderate	Not unique; see §6.14
Delusion / completeness not “more”	Retract not iso	Strong	Coherent and categorical
Graded values (D4)	Enrichment over preorder `W`	Moderate	Compresses full taxonomy
Evaluation μ, justice operator	`Eval` on `Liv`; partial composite $\rho \to \varphi \to \mu$	Moderate	Order-restriction; φ fibred over cap (§6.6.1)
cap(u) triad (D5a)	Fibred enrichment; `cap : Liv → Cap`	Moderate	Preorders given; frustration = failed composition
Right-use vs consumption	Natural transformation	Strong	Naturality square is real constraint
Unit relationships	`Rel`, functor `V : Rel -> Val`	Moderate	Effective φ_u family, not global functor
Undivided society	Colimit + compatibility (L4)	Strong	Compound κ_comp reading of §6.8
Mixture vs compound (D8)	Coproduct vs fusion colimit	Moderate	Modes distinguished §6.9; content guard separate
Transmission τ (L5)	`Trans` coalgebra (sketch)	Weak	Not derivable from κ

Madhyasth concept	Categorical representation	Fit	Comment
Complementarity (L3)	`AdmissibleComp` guard on spans	Partial	Predicate on diagrams, not CT-derived (§6.9.1)
Development / effort	Monoidal/Petri; κ_comp transitions	Moderate	*vikas-kram* bookkeeping only
Education / sanskar	τ + faithfulness of Info→Conduct	Moderate	Tied to transmission
Means cannot fix ends	No generating functor (Hume)	Weak as novelty	True; philosophy supplies it

6.13 What does NOT fit well (and why)

1. **`Jeevan` as a substance.** Category theory is structuralist: by Yoneda, an object just *is* its web of morphisms. A faithful categorical reading inevitably re-describes `jeevan` as a **functional role** — precisely the reductionist/functionalist position the darshan rejects. The formalism cannot represent "substantial existence over and above relational role" without leaving category theory (What Is Existence? §6.6 Saturation-Reflector).
2. **Samadhi / samyama as the warrant.** Nagraj's epistemic foundation is realisation through `sadhana-samadhi-samyama` (MVD, point 4): private, first-person, non-relational. It is the *source of the axioms* and necessarily sits **outside** the category.
3. **Constitutional completeness as content.** Encodable as an irreversible transition, but the Petri structure is content-agnostic; the specific metaphysical claim is invisible to it.
4. **Saturation vs unit relationships.** Omnipresence energises and regulates every unit (A1–A2) but is **not** a member of **U** and not an arrow in `Rel`. Saturation is the first ontological layer; *sambandh* is the second (What Is Existence? §1.2.1). Terminal object, monoidal unit, or topos-as-space are inadequate substitutes.
5. **Coexistence: ground and constraint.** It functions as the background that makes composition possible and as a constraint (colimit compatibility). That dual role signals a **first principle** not fully internalisable as a single categorical object.
6. **Truth/realisation-in-existence (`satya`).** Temptingly a terminal object (unique target), but realisation is experiential and self-involving, which the terminal-object analogy loses.
7. **The self-knowing of the knowledge order.** Knower and known coincide — reflexive. This gestures at fixed points or Yoneda self-reference but risks paradox and is not obviously well-founded.

8. **Normativity (`ought`).** Category theory describes structure, not obligation. Arrows like `usesRightly` smuggle an "ought" into a descriptive arrow; the is/ought gap is untouched.
9. **Complementarity as selection (L3).** Colimits and coproducts describe **how** to glue; L3 specifies **which** spans fire. The guard `AdmissibleComp` (§6.9.1) must be imported as data — P9 and P12.
10. **Order-specific cap preorders.** The fibred model (§6.6.1) types ksh, yog, pat by order, but the numeric or comparative structure of those preorders is textual, not categorical.

6.14 Issues and open problems

1. **Non-uniqueness of the formalization.** Body/ `jeevan` could be a monoid action, a fibration, a comma object, or a monad algebra; cap(u) could be a fibration, a profunctor, or a weighted enrichment — the "theory" is really a *family* of models.
2. **The premises do the work.** In all propositions except P6, the categorical step is valid but inert without a contested premise, or requires structure outside CT (P8–P9, L3 guard).
3. **Structuralism vs substantialism** (§6.13.1) is unresolved and possibly unresolvable within category theory — a boundary of the tool, not a bug.
4. **No empirical contact.** Nothing yields a measurement or prediction of `jeevan`, coexistence, or constitutional completeness.
5. **Risk of mathematical theater.** Elegant diagrams can create an illusion of proof. The conditional framing and this list are the safeguard.
6. **Enrichment base underdetermined.** The quality preorder `W`, D4 taxonomy, and cap preorders are chosen, not derived.
7. **Saturation and τ resist full internalization** (§6.13.4–5, P8).
8. **L3 guard underdetermined.** `AdmissibleComp` is named and typed (§6.9.1) but not uniquely specified — the same span may be admissible under complementarity yet fail other textual constraints (proportion, order).
9. **Mixture/compound boundary cases.** Real assemblies may exhibit intermediate behaviour (partial fusion). The binary κ_mix / κ_comp split is a modelling convenience; finer quotients may be needed for some tiers.

7. Limits and conclusion

A clear guide must also say what this does **not** do.

- **It does not prove the philosophy.** Drawing `jeevan` as a box does not show `jeevan` exists. The diagrams organise the claims; they do not verify them.
- **It can quietly shrink `jeevan`.** Category theory describes things purely by their relationships. But the darshan insists `jeevan` is a real *entity*, not just a role it plays. The more faithfully you use the mathematics, the more it risks turning `jeevan` into "the thing that does these jobs" — closer to the materialist view the darshan rejects. This tension is real and unresolved.
- **It cannot capture realisation.** The darshan's foundation is a first-person, meditative realisation (`samadhi`). That is the *source* of the assumptions, and it sits outside any diagram.
- **It adds no evidence.** No measurement of `jeevan`, constitutional completeness, or coexistence comes out of this. For that, science would still need testable definitions and observations.

Category theory is a **lens for clarity**, not a proof machine and not a new physics. It makes Madhyasth Darshan's logic visible, shows exactly what each conclusion depends on, maps naturally onto much of [The Coexistence Template]() — including fibred **cap(u)** (§6.6.1) and **mixture/compound** κ (§6.9) — and contributes one real insight about social gluing, while leaving saturation, the L3 guard, transmission, and substantial *jeevan* as explicit boundaries.

```
flowchart LR
    Co["Coexistence"] --> Un["Understanding"]
    Un --> Re["Resolution"]
    Re --> Hu["Humane conduct"]
    Hu --> Ri["Right-use"]
    Ri --> So["Undivided society"]
```


References

Madhyasth Darshan (primary sources)

- **MVD** — Nagraj, A. *Madhyasth Darshan — Co-existentialism*. English translation by Rakesh Gupta. Cited: four orders; relationships and values; capacity–ability–receptivity (pp. 62, 79, 134, 142); mixture and compound (p. 42); hungry/overfull atoms (p. 8); three satisfactions (Ch. 4); organisation sustainment (p. 55).
- **SB** — Nagraj, A. *Samadhanatmak Bhautikvad / Resolution Centred Materialism*. English translation by Rakesh Gupta. Cited: effort–motion–result; constitutional completeness (p. 55); value-distinct acts (Ch. 7).
- **JV** — Nagraj, A. *Jeevan Vidya: An Introduction*. English translation by Rakesh Gupta. Cited: body as instrument (Ch. 1); complementarity (p. 157); commerce and "more" (p. 41).

Related studies in this collection

- *What Is Existence?* — ontological exposition: saturation, *sambandh*, *gathanpurnata*, Development and Awakening Progression, activity triad
- *The Coexistence Template* — formal template (κ, τ, ρ, φ, μ, L1–L6); reciprocal link at §7
- *Nature of Time* — *kaal* as duration of unit-activity; *shram–gati–parinam* and directional *vikas*
- *Why Humans Are Not Just Material* — human-tier anthropology
- *Human Behavior and Society* — conduct and social order